Exercice 1

$\underline{t<0}$ $u_e(t)=0$, s(t) $=0$

$\underline{t>0}$

$$u_s(t)+\frac{1}{\omega_c}\frac{du_s}{dt}=-2u_e(t)$$

$$u_s(t)=-2(1-e^{-\frac{t}{\tau}})$$

Exercice 2

$$t>0$$

$$u_s(t)+\frac{1}{Q\omega_0}\frac{du_s}{dt}+\frac{1}{\omega_0^2}\frac{d^2u_s}{dt^2}=-\frac{5}{\omega_0^2}\frac{d^2u_e(t)}{dt^2}=0$$

$$\frac{d^2u_s}{dt^2}+\frac{\omega_0}{Q}\frac{du_s}{dt}+\omega_0^2u_s(t)=0, Q=4$$

$$\Delta=\omega_0^2(\frac{1}{Q}-4)<0$$

$$r_1^2=-\frac{\omega_0}{2Q}\pm j\omega_0\sqrt{1-\frac{1}{4Q^2}}$$

$$u_s(t)=ae^{r_1t}+be^{r_2t}$$

$$=e^{-\frac{\omega_0}{2Q}t}(Acos(\omega_0t+\varphi))$$

$$u_s(0^+)-u_s(0^-)=\underline{H}(\omega\to+\infty)(u_e(0^+)-u_e(0^-))$$

$$\int_{0^-}^{0^+}u_s(t)+\int_{0^-}^{0^+}\frac{1}{Q\omega_0}\frac{du_s}{dt}+\int_{0^-}^{0^+}\frac{1}{\omega_0^2}\frac{d^2u_s}{dt^2}=\int_{0^-}^{0^+}-\frac{5}{\omega_0^2}\frac{d^2u_e(t)}{dt^2}$$

$\begin{cases}u_s(0^+)=-5\\ \frac{du_s(0^+)}{dt}=\frac{5}{4}\omega_0\end{cases}$

$A^{=}25+\frac{25}{64}$

$tan\varphi=\frac{1}{8}$

$A\simeq-5$

$$u_s(t)=-5e^{-\frac{\omega_0t}{8}}cos(\omega_0t+arctan(\frac{1}{8}))$$

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$r_1^2=-\frac{-\omega_0}{0.4}\pm\frac{\omega_0\sqrt{21}}{2}$

$\nearrow -0.2\omega_0$

$\searrow -4.8\omega_0$

$$u_s(t)=\lambda e^{-0.2\omega_0t}+\mu e^{-4.8\omega_0t}$$

$u_s(0^+)=-5$

$\frac{du_s}{dt}(0^+)=25\omega_0$

$\begin{cases}\lambda\simeq0.21\\ \mu\simeq-5.21\end{cases}$

$$u_s(t)=0.21e^{-0.2\omega_0t}-5.21e^{-4.8\omega_0t}$$

4/ Q=4 résonnance $\omega_c=\omega_0\sqrt{1-\frac{1}{2Q^2}}$

$G_dB=20log5+40log(\frac{\omega}{\omega_0})$